

Problem C. Make it Palindrome 2

Time Limit 2000 ms

Problem Statement

You are given a positive integer N , a positive integer M , and an integer sequence $A = (A_1, A_2, \dots, A_N)$ of length N . It is guaranteed that each element of A is between 0 and $M - 1$, inclusive.

You can perform the following operation on the integer sequence A any number of times, possibly zero:

- Choose a pair of integers (l, r) satisfying $1 \leq l \leq r \leq N$, and replace A_i with $(A_i + 1) \bmod M$ for each $i = l, l + 1, \dots, r$.

Find the minimum number of operations required to make A a palindrome.

Here, A is a palindrome if $A_i = A_{N+1-i}$ holds for $i = 1, 2, \dots, N$.

You are given T test cases; solve each of them.

Constraints

- $1 \leq T$
- $1 \leq N \leq 2 \times 10^5$
- $1 \leq M \leq 10^9$
- $0 \leq A_i < M$
- The sum of N over all test cases is at most 2×10^5 .
- All input values are integers.

Input

The input is given from Standard Input in the following format:

```
T
case1
case2
⋮
caseT
```

Each test case is given in the following format:

$$N \ M$$

$$A_1 \ A_2 \ \dots \ A_N$$

Output

Output the answers for the test cases in order, separated by newlines.

Sample 1

Input	Output
3 4 5 0 3 1 2 1 20260418 454 7 12 3 1 4 1 5 9 2	3 0 5

Consider the first test case.

You can make A a palindrome in three operations as follows:

- Choose $(l, r) = (2, 4)$. A becomes $(0, 4, 2, 3)$.
- Choose $(l, r) = (3, 4)$. A becomes $(0, 4, 3, 4)$.
- Choose $(l, r) = (3, 4)$. A becomes $(0, 4, 4, 0)$.

It is impossible to make A a palindrome in fewer than three operations, so output 3.